

File permissions in Linux

Project description

In the company there is a team called Research Team. They currently need to confirm the permissions given on the `projects` directory so it can be updated and make sure it has the permissions and/or restrictions required for security and reflect the level of authorizations that should be given. I will address the situation by certain tasks that you will see next.

Check file and directory details

Normally we can use `ls` to show what is in the folder, but if we want to show permissions for directories, subdirectories and files in the current directory, we have to add `-la`, so the command is `ls -la`.

```
researcher2@14abd0ab442b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Feb  4 01:21 .
drwxr-xr-x 3 researcher2 research_team 4096 Feb  4 02:20 ..
-rw--w---- 1 researcher2 research_team   46 Feb  4 01:21 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Feb  4 01:21 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Feb  4 01:21 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Feb  4 01:21 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Feb  4 01:21 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Feb  4 01:21 project_t.txt
researcher2@14abd0ab442b:~/projects$
```

From here at the beginning of each line we can see the first 10 characters meaning permissions. Also we can see the name of the files at the end of the lines, including hidden files such as `.project_x.txt` which can be identified by a hyphen (.) at the beginning of their names.

Describe the permissions string

The first 10 characters mean the permissions so to be clear it can be explained in this way:

1st character: If it is a `d`, it is a directory. If it is a `-`, is a file.

2nd, 3rd and 4th characters: Shows the permissions for the `user` role. Where if we find a `r` is for read, `w` for write and `x` for execute. The order is normally `rw`, but if we find instead a `-`, is for no permission granted.

5th, 6th and 7th characters: Shows the permissions for the `group` role. Where if we find a `r` is for read, `w` for write and `x` for execute. The order is normally `rw`, but if we find instead a `-`, is for no permission granted.

8th, 9th and 10th characters: Shows the permissions for the `other` role. Where if we find a `r` is for read, `w` for write and `x` for execute. The order is normally `rw`, but if we find instead a `-`, is for no permission granted.

Change file permissions

The Research Team indicates that does not want others to have write access to any file. I confirmed that only file with allowing write permissions to others is `project_k.txt`, so I proceeded by using the `chmod` command to make the change.

```
researcher2@14abd0ab442b:~/projects$ chmod o-w project_k.txt
researcher2@14abd0ab442b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Feb  4 01:21 .
drwxr-xr-x 3 researcher2 research_team 4096 Feb  4 02:20 ..
-rw--w---- 1 researcher2 research_team  46 Feb  4 01:21 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Feb  4 01:21 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Feb  4 01:21 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Feb  4 01:21 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb  4 01:21 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb  4 01:21 project_t.txt
researcher2@14abd0ab442b:~/projects$
```

If we want to change permissions, we have to follow the next order:

- `chmod` (to ask change the permission)
 - the first letter of the role we want to change the permissions for, with an `+` to add permissions or `-` to remove permissions, and the permission we want to add/remove
 - the directory or file we want to edit the permissions for

So, in this scenario we used the command `chmod o-w project_k.txt`, after that using `ls -la` we confirmed that we got success result of removing (`-`) the write (`w`) permission to the other (`o`) role of the file `project_k.txt`.

Change file permissions on a hidden file

```
researcher2@14abd0ab442b:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@14abd0ab442b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Feb  4 01:21 .
drwxr-xr-x 3 researcher2 research_team 4096 Feb  4 02:20 ..
-r--r----- 1 researcher2 research_team  46 Feb  4 01:21 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Feb  4 01:21 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Feb  4 01:21 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Feb  4 01:21 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb  4 01:21 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb  4 01:21 project_t.txt
researcher2@14abd0ab442b:~/projects$
```

Same process as previous example, but the difference in this one is that we are required to change various permissions at the same time, and that can be accomplished by separating the requests by a comma (,) with no spaces between. For this one we removed (-) write (w) permissions for users (u) and groups (g) and added (+) read (r) permission to groups (g). Example: `chmod u-w,g-w,g+r .project_x.txt`

Change directory permissions

```
researcher2@14abd0ab442b:~/projects$ chmod g-x drafts
researcher2@14abd0ab442b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Feb  4 01:21 .
drwxr-xr-x 3 researcher2 research_team 4096 Feb  4 02:20 ..
-r--r----- 1 researcher2 research_team  46 Feb  4 01:21 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Feb  4 01:21 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Feb  4 01:21 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Feb  4 01:21 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb  4 01:21 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb  4 01:21 project_t.txt
researcher2@14abd0ab442b:~/projects$
```

Same process as previous example, but the difference in this one is that we are required to remove (-) execute (x) permission to the group (g) role on the subdirectory `drafts`. Example: `chmod g-x drafts`

Summary

I had to make multiple changes on permissions to match the requirements by the Research Team, making sure we achieved the level of authorizations desired by them on the `projects` directory. Is important to use `ls -la` since it allows us to confirm we are making the changes properly and use the `chmod` command for each file or directory we want to manage.

